NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

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13. ABSTRACT

This Engineering Test Procedure describes test methods and techniques for evaluating the technical performance and characteristics of Collective Protectors (Fixed-Installation Type) (Chemical, Biological, Radiological). The evaluation is related to criteria expressed in applicable Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), or other design requirements and specifications.

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2 March 1970

U. S. ARMY TEST AND EVALUATION COMMAND
COMMODITY ENGINEERING TEST PROCEDURE

COLLECTIVE PROTECTORS, FIXED-INSTALLATION

1. OBJECTIVE

The objective of this commodity engineering Materiel Test Procedure (MTP) is to establish uniform procedures for determining and evaluating the technical performance and safety aspects of fixed-installation collective protectors in terms of the criteria established by applicable Qualitative Materiel Requirements (QMR), Small Development Requirements (SDR), Technical Characteristics (TC), and other design requirements or specifications.

2. BACKGROUND

A fixed-installation collective protector is a device intended to allow personnel to live and work in a permanent structure during and after an attack with chemical, biological, or radioactive (CBR) agents without wearing individual protective equipment.

A fixed-installation collective protector delivers filtered air to a building such as a housing unit, barracks, hospital, office or shop to ventilate the building while protecting personnel against CBR agents in the surrounding atmosphere.

A collective protector must protect personnel against CBR agents in both particulate and nonparticulate (vapor) form. Accordingly, any such device will contain means of removing both dusts and gases from air and of circulating that air into the protected building. Collective protectors may incorporate means for heating and/or cooling or otherwise conditioning the circulated air, means for selecting or proportioning between outside air and recirculated air, provisions for using particulate and gas filters individually or in series, and controls for regulating the air flow to meet varying demands for conditioning and replacement of the air in the enclosed building. Collective protectors may be operated by electric motors or by other power such as gasoline engines.

Fixed-installation protectors are life-support equipment; it is therefore essential to measure and evaluate their technical performance.

3. REQUIRED EQUIPMENT

a. Environmental Chambers for:

1) Temperature testing
2) Fungus testing
3) Humidity testing
4) Dust testing
5) Water testing
6) Salt fog testing
7) Rain testing

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b. Suitable Test Site
c. Photographic Equipment, Still and Motion Picture Cameras
d. Meteorological Equipment to measure and record:

1) Temperature
2) Wind direction and speed
3) Relative humidity

e. Protective Clothing
f. Aircraft or Simulated Aircraft
g. Accelerometers, as required
h. Transportability testing facility for:

1) Vibration
2) Shock

i. Materials Handling Equipment
j. Leak Testing Facilities
k. Various Types of Test Gases and method of gas dispersion

4. REFERENCES

A. USATECOM Regulation 385-6, Verification of Safety of Materiel During Testing.
B. USAMC Pamphlet 706-134, Engineering Design Handbook: Maintainability Guide for Design.
C. MIL-STD-810B, Environmental Test Methods.
D. MIL-STD-1472, Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
E. MIL-H-46855, Human Engineering Requirements for Military Systems, Equipment, and Facilities.
F. MTP 7-1-002, Air Portability and Airdrop Service Testing.
G. MTP 7-2-509, Airdrop Capability of Materiel.
H. MTP 7-2-515, Air Transport (Suitability of Equipment for).
I. MTP 8-2-192, Collective Protection Systems, Vehicles and Vans.
J. MTP 8-2-193, Collective Protection Systems, Field Shelters.
K. MTP 8-2-500, Receipt Inspection.
L. MTP 8-2-503, Rough Handling and Surface Transport.
M. MTP 8-2-510, Decontamination.
N. MTP 8-2-511, Leak Testing of Protective Equipment.

5. SCOPE

5.1 SUMMARY

The procedures required in this MTP are divided into a series of subtests. The sequence will be determined by the test plan.

The following subtests comprise the complete procedures:

a. Receipt Inspection - An inspection of the test item as received

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to (1) determine its physical characteristics and condition, (2) locate any defects it might have, and (3) identify damage received during transport. During this inspection the test items will also be serially numbered for subsequent identification.

b. Installation - A description of the test set-up to be used and an evaluation of the installation procedures.

c. Safety Evaluation - The objective of this procedure is to check the safety statement issued by the developing agency and to identify the safety hazards, if any, which must be included in the safety release recommendation required by USATECOM Regulation 385-6.

d. Simulated Environmental Testing - An evaluation to determine the effects of extreme temperatures, fungus, humidity, dust, and fresh and salt water on the test item.

e. Rough Handling and Surface Transport - An evaluation to determine the effects of rough handling and surface transport on the physical and operational characteristics of the test item.

f. Air Transportability - An evaluation to determine the effects of air transport conditions on the physical and operational characteristics of the test item.

g. Airdrop Capability - An evaluation to determine the effects on the test item of being subjected to airdrop conditions.

h. Leak Testing - An evaluation to determine if the test item leaks when subjected to standard leak tests and conditions.

i. Filter Tests - An evaluation to determine if the filters which are part of the test item satisfy the design requirements.

j. Operational Reliability - An evaluation to determine if the test item, installed in a simulated protective facility, meets specified reliability criteria.

k. Decontamination Aspects - An evaluation to determine the ease with which the test item can be decontaminated and the effects of the process on the item.

l. Maintenance Characteristics - An evaluation to determine the ease of performing the required maintenance on the test item and the need for special tools and skills.

m. Human Factors Aspects - An evaluation to assess the ease of transporting, installing, and operating the test item.

5.2 LIMITATIONS

This MTP is not intended to provide procedures for collective protection systems or for collective protectors designed for use in vehicles, vans, field shelters since these commodities are covered in MTP 8-2-192 and 8-2-193. The testing of heating and cooling devices which may be incorporated in a collective protection system is not included since the testing of such sub-assemblies are subject to other MTP's.

6. PROCEDURES

6.1 PREPARATION FOR TEST

6.1.1 Safety Statement

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The test officer will insure that a Safety Statement is received from the developing agency before commencing testing and that it is understood by all test personnel. The Safety Statement includes information pertaining to the test item's operational limitations and specifies hazards peculiar to the item or components which are to be tested.

6.1.2 Safety Procedures

a. Test plans and procedures will insure performance in the safest manner consistent with accomplishing the mission. Plans will include safety procedures, precautions, protections, and emergency procedures as necessary. Technical information on the hazards and safety characteristics of the test item as provided by the safety statement and other pertinent information will be included. Such information will include an evaluation of potential hazards, analysis of risks, limitations, and precautions, including special test equipment and techniques, that should be incorporated in test plans and procedures.

b. One individual will be charged with responsibility for safety. He will be familiar with the construction and operation of the test item and its critical components, have full knowledge and the hazards and safety aspects of the test, review test procedures for evaluation of hazards, and recommend control measures.

c. All personnel who participate in or observe the tests will be briefed on the proper test methods and procedures.

d. When dealing with chemical and biological agents, all test personnel must constantly consider not only hazards that may be encountered during normal conditons, but also those which could be encountered under the worst conditions of malfunctioning, accidents, or emergencies.

e. Test personnel must be able to use chemical agent detector kits, recognize symptoms of chemical agent poisoning in themselves and others, and use first-aid and decontaminating procedures.

f. A record will be kept of injuries suffered by test personnel during testing, regardless of how minor they may be and regardless of their relevance to testing.

6.1.3 Security

Security considerations will be provided for as applicable to each of the procedures described in this MTP.

6.1.4 Logistical Requirements

Prior to the conduct of any subtest, the test officer will insure that all logistical requirements are satisfied.

6.1.5 Test Sequence and Sample Size

The procedures required in this MTP are divided into a series of subtests. The sequence will be determined by the test plan. Operational tests are divided into two groups: (1) filter tests, intended to evaluate the performance of filters, and (2) simulated installation tests, intended to evaluate the test item operation when installed in a fixed protective installation. In preparing the test plan, consideration should be given to the number of test items available, their susceptibility to damage, time available, availability

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of facilities, reliability and confidence limits set by QMR and SDR, and budget limitations. Subtests deemed most likely to cause failure should be performed first when possible so that the developing agency may have the earliest possible notice of the deficiency.

6.2 TEST CONDUCT

6.2.1 Receipt Inspection

The test item will be subjected to the applicable procedures of MTP 8-2-500 following its arrival at the test site, with emphasis on the following:

a. Determine the adequacy of packaging by visually inspecting the test item crates and packages. Record the following:

1) Binding deficiencies such as broken straps, seals, etc.
2) Packaging material deficiencies such as cuts, tears, breaks, etc.
3) Rust and corrosion of metal.
4) Illegible or missing markings.
5) Presence of packing lists and uncrating instructions.
6) Missing packages.

b. Test item inspection:

1) Uncrate the test item and note adequacy of instructions and special tools required.
2) Visually inspect the test item for broken components, dents, cracks, other damage, illegible markings, etc.
3) Ascertain that the proper type, size, and number of filters are present.

c. Determine and record the dimensions and weights of each crated and uncrated test item component.

d. Number serially and identify each test item component to be used.

e. Photograph all defective test items.

6.2.2 Installation

Install the test item in a simulated fixed protective installation as appropriate for each test to be performed:

6.2.2.1 Filter Test Setup

a. The filter unit will be mounted in a duct whose length is equal to at least ten times the largest dimension of the filter.

b. The duct will be properly instrumented to record the gas flow through and the pressure drop across the filter.

c. The duct will include straighteners or will be otherwise designed to minimize the velocity variation across the area of the filter.

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d. Record missing components, improper fits, and difficult assembly operations.

6.2.2.2 Collective Protector Test Setup

Those test procedures which involve testing of the entire collective protector will be performed in a simulate fixed installation. The setup is as follows:

a. In a suitable test area, erect a simulated, typical, fixed collective protection installation. Install the protector as specified in the technical manual, or as specified by the test officer. Record ease of installation.

b. The volume and area of the simulated collective protection installation will be as specified in the test item requirements.

c. The simulated installation will be erected next to or as close as possible to an environmental test chamber so that the test chamber may be used to simulate adverse weather conditions and to contain CBR challenges of the collective protector as specified in paragraph 6.2.10.

6.2.3 Safety Evaluation

a. Observe the installation and operation of the equipment in accordance with existing instructions, instruction manuals, directives, safety SOP's, and similar guidance. Record any hazardous conditions and operations.

b. Record information for inclusion in the Safety Release required by USATECOM Regulation 385-6.

c. Perform additional checks as required to verify all the safety aspects included in the Safety Statement prepared by the developing agency. Record deficiencies and recommended inclusions.

6.2.4 Simulated Environmental Testing

Unless the test item requirements or the test officer directs otherwise, a representative number of filters and at least one simulated fixed-installation protective collector will be subjected to the following simulated environmental tests:

6.2.4.1 Cyclic Storage

a. Subject the test item in its packing container to cycles of climatic extremes. A cycle shall consist of three weeks duration as follows: successive one week tests at humid, low temperature, and high temperature. Chamber conditions for each climatic condition are as follows:

1) Humid Storage. The chamber shall be maintained at 113°F ± 2°F and 85% R.H. for the duration of the test.
2) Low Temperature Storage. The chamber shall be maintained at -65°F ± 2°F for the duration of the test.
3) High Temperature Storage. The chamber shall be maintained

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at 160° F ± 2° F for the duration of the test.

b. The test item shall be subjected to a minimum of three such cycles, or more if specified. Upon completion of each cycle, the container and contents shall be examined for damage.

6.2.4.2 Extreme-Temperature Tests

6.2.4.2.1 Low-Temperature Tests - Perform the following:

a. Subject a minimum of 4 test items to a temperature of -53.9° C (-65° F) for a period of 72 hours, and visually inspect the test items and record damage.

b. Subject the test items to the minimum operating temperature as established by design requirements, and maintain this temperature until stabilization is reached. If stabilization is attained in less than 24 hours, maintain temperature for complete 24-hour interval. Perform the following:

NOTE: Stabilization, unless otherwise specified, is considered to be reached when the temperature of the test item does not change more than 2° C (3.6° F) per hour.

1) Visually inspect the test items, and record damage.
2) Subject 1/4 of the test items to the applicable leaks tests of paragraph 6.2.8.
3) Subject 1/4 of the test items to the applicable operational tests of paragraphs 6.2.9 and 6.2.10.

c. Allow the temperature of the test item to stabilize at local ambient conditions, and perform the following:

1) Visually inspect the test item and record damage.
2) Subject 1/2 of the remaining test items to the applicable leak test procedures of paragraph 6.2.8.
3) Verify the operability of the remaining test items by subjecting them to the applicable procedures of paragraph 6.2.9 and 6.2.10.

6.2.4.2.2 High-Temperature Tests - Perform the following:

a. Subject a minimum of 8 test items to a temperature of 68.3° C (155° F) and a relative humidity of 15 percent. Maintain these conditions for a minimum of 4 hours, and visually inspect the test item and record damage.

b. Subject the test items to a temperature of 48.9° C (120° F) and a relative humidity of no more than 15 percent, and maintain these conditions for a minimum of 24 hours. Then perform the following:

1) Visually inspect the test item and record damage.
2) Subject 1/4 of the test items to the applicable leak test procedures of paragraph 6.2.8.

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3) Subject 1/4 of the test items to the applicable operational procedures of paragraphs 6.2.9 and 6.2.10.

c. Subject the test items to local ambient temperatures and humidity for 24 hours, and perform the following:

1) Visually inspect the test item and record damage.
2) Subject ½ of the remaining test items to the applicable leak test procedures of paragraph 6.2.8.
3) Subject the remaining test items to the applicable operational procedures of paragraphs 6.2.9 and 6.2.10.

6.2.4.3 Fungus Test

a. Subject a minimum of 2 test items to the fungus test of Procedure I, Method 508, reference 4C (MIL-STD-810B).
b. At the completion of the cycling period, perform the following:

1) Visually inspect the item and its components, and record the presence of fungus or corrosion.
2) Subject the test items to the applicable operational test procedures of paragraph 6.2.10.

6.2.4.4 Humidity Test

a. Subject a minimum of 2 test items to the humidity cycling of Procedure I, Method 507, reference 4C (MIL-STD-810B).
b. At the completion of the cycling period, perform the following:

1) Visually inspect the item and record signs of corrosion.
2) Subject the test items to the applicable procedures of paragraphs 6.2.9 and 6.2.10.

6.2.4.5 Dust Test

a. Subject a minimum of 2 test items to the dust exposure conditions of Procedure I, Method 510, reference 4C (MIL-STD-810B).
b. At the completion of the exposure period, perform the following:

1) Visually inspect the test items and record damage.
2) Verify the operability of the test item by subjecting them to the applicable procedures of paragraphs 6.2.9 and 6.2.10.

6.2.4.6 Water Immersion Tests

If the item design requirements require it, perform the following:

a. Subject a suitably determined number of test item components, in their packing cases, to the water immersion test of Procedure I, Method 512, reference 4C (MIL-STD-810B).

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NOTE: If design requirements establish depth of water, water temperature, or time of immersion different from the standard procedure, the test plan will so state.

b. At the completion of the immersion test, remove the test item components from their containers, and perform the following:

1) Inspect the components for evidence of water damage.
2) Subject the test item components to the operability tests of paragraphs 6.2.9 and 6.2.10, if applicable.

6.2.4.7 Salt Fog Test

a. Subject a minimum of 2 test items to the conditions of Procedure I, Method 509, reference 4C (MIL-STD-810B).

b. At the completion of the salt fog spray exposure, perform the following:

1) Rinse the test item with clear water, if applicable.
2) Visually inspect the test item and its components for the presence of corrosion.
3) Subject ½ of the test items to the applicable leakage test of paragraph 6.2.8.
4) Verify the operability of the remaining test items by subjecting them to the applicable procedures of paragraphs 6.2.9 and 6.2.10.

6.2.4.8 Rain Test

a. Subject a minimum of 2 test items to the rain conditions of Procedure I, Method 506, reference 4C (MIL-STD-810B).

b. At the completion of the rain exposure, perform the following:

1) Visually inspect the test items and its components and record the following:

a) Evidence of water penetration
b) Presence of corrosion or other damage

2) Verify the operability of the test items by subjecting them to the applicable procedures of paragraphs 6.2.9 and 6.2.10.

6.2.5 Rough Handling and Surface Transport Tests

a. Subject the test item components, packaged in their original containers, to the following procedures of MTP 8-2-503.

1) The vibration test of paragraph 6.2.2.2a.3
2) The shock test of paragraph 6.2.2.1a.2

b. At the completion of testing, perform the following:

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1) Examine the packaging for cracks, breaks, undone binding, etc.
2) Examine the test item components for damage and deformation.
3) If required by the test item design requirements, assemble the components on a fixed installation according to paragraph 6.2.2 and subject the unit to the applicable leak tests of paragraph 6.2.8.
4) Verify the operability of the test item by subjecting it to the procedures of paragraph 6.2.10.

6.2.6 Air Transportability

If required by the test item's design requirements, determine the ease of loading and unloading the test item components from an aircraft as described in the applicable portions of MTP 7-2-515 or as follows:

NOTE: Background information on air transportability is contained in MTP 7-1-002.

a. Load the test item components, in their shipping containers, aboard a typical cargo aircraft or simulated aircraft, using current standard loading equipment, and record the following:

1) Type of aircraft used or simulated
2) Dimensions, weight and material of shipping containers
3) Equipment used for loading
4) Difficulties encountered while loading
5) Method of tiedown
6) Any damage sustained by the packages during loading

b. Unload the test item components from the aircraft or simulated aircraft, and record the following:

1) Equipment used in unloading
2) Difficulties encountered while unloading
3) Damage sustained during unloading

6.2.7 Airdrop Capability

If required by the test item specifications, subject a minimum of 2 test item components, packaged in their original containers, to the applicable sections of MTP 7-2-509 and as follows:

a. Rig the containers, with attached accelerometers, in appropriate airdrop containers, and drop them from typical aircraft as instructed by MTP 7-2-509. Record the following:

1) Aircraft type(s) used
2) Aircraft airspeed
3) Altitude
4) Meteorological conditions
5) Impact velocities

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6) Deceleration magnitude at impact in g's

b. Cover the airdrop test procedures with still and motion cameras.
c. At completion of the test, perform the following:

1) Examine the test item's packaging for breaks, undone bindings, etc.
2) Examine the test item components for damage and deformation.
3) Subject ½ of the test items to the leak test of paragraph 6.2.8.
4) Subject the remainder of the items under test to the operability procedures of paragraphs 6.2.9 and 6.2.10.

6.2.8 Leak Testing

Subject the test item to the leak test procedures of MTP 8-2-511 to determine whether the filter edges will seal around their periphery when installed in the collective protector.

6.2.9 Filter Tests

6.2.9.1 Flow-Rate Test

a. Subject the particulate and gas filters to the following tests or as revised by the test plan:

1) Install the particulate filter as specified in paragraph 6.2.2.1. Conduct the test at the constant flow rate specified by the equipment specifications. Determine the pressure drop across the filter and the airflow rate.
2) Install the gas filter as specified in paragraph 6.2.2.1 and adjust the airflow rate according to the equipment specifications. Measure the pressure drop across the filters and the flow rate.

b. Record the following data:

1) Test item identification (filter serial numbers)
2) Airflow rate
3) Pressure drop across filter
4) Testing difficulties, if any

6.2.9.2 Dioctyl Phthalate (DOP) Filter Penetration Test

Particulate filters should provide protection against CBR agents in the particulate form. The following test will be conducted to determine whether the filter will provide this protection.

a. Perform the following:

1) Mount particulate filter as described in paragraph 6.2.2.1.

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2) Subject the filter to airflow containing a specified concentration of dioctyl phthalate of known particle size range.
3) Sample the air leaving the exit for DOP penetration.

b. Record the following data:

1) Test item identification.
2) Type, size, and weight of filter being tested.
3) Flow rate of air.
4) Pressure drop across filter.
5) Results of air sampling test at exit of filter unit, which should determine the smoke density in the purified air (percent DOP penetration).
6) Reasons for difference in performance from those specified in the QMR, SDR, or other technical specifications.
7) DOP particle size used.

6.2.9.3 Filter Gas Life Test

a. Evaluate the degree and time duration of protection afforded by the gas filters by performing the following:

1) Weigh the filter.
2) Install filters as described in paragraph 6.2.2.1.
3) Admit air mixed with a specific proportion of phosgene or similar agent at the filter's allowable flow rate. Gas concentration will be specified by QMR, SDR, or other technical specifications.
4) Determine the flow resistance across the filters and the airflow.
5) Measure the concentration of gas passing through the filter.
6) When the gas passing through the filter increases to the value specified in the requirements, stop the test.
7) Reweigh the filter.

b. Record the following:

1) Test item identification
2) Airflow rate
3) Flow resistance of filters (inches of water)
4) Type and concentration of agent used
5) Filter life
6) Weight of filters before and after test

6.2.10 Operational Reliability

The following tests will be performed with the test item installed in a simulated fixed collective protection installation as described in paragraph 6.2.2.2.

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6.2.10.1 Flow-Rate Test

a. Operate the test item with 100 percent fresh, unfiltered air and also with various proportions of recirculated air as specified in the test plan.

b. Measure and record the flow rates and the static pressure in the shelter.

c. Measure and record the stall pressure of the test item, i.e., the static pressure in the shelter at which no more air is admittted.

6.2.10.2 Agent Challenge Tests

The following tests will determine the ability of the test item to protect an installation from challenges by various materials:

6.2.10.2.1 Dioctyl Phthalate (DOP) Challenge - Envelop the collective protector intake completely with DOP smoke. Draw air into the protective installation at the rate specified by the test item requirements or the test plan. Perform the following:

a. Measure and record the rate of DOP penetration of the collective protector.

b. Measure and record the DOP concentration in the protective installation as a function of time.

c. Compare the data obtained in this test with those obtained when the filters were tested alone (paragraph 6.2.9.2).

d. Measure and record the airflow rate and pressure in the installation.

6.2.10.2.2 Sarin (GB) Challenge - Perform the following:

a. Challenge the protective installation air inlets with sarin in a manner which will subject the test item to a reasonable amount of vapor contamination.

NOTE: The test plan or the design requirements will determine what is reasonable.

b. Repeat steps a through d of paragraph 6.2.10.2.1.

6.2.10.2.3 Distilled Mustard Gas (HD) Challenge - Repeat test procedure of paragraph 6.2.10.2.1 using HD agent.

6.2.10.2.4 Cyanogen Chloride (CK) and Sarin (GB) Challenge - Subject the protector inlet to a CK challenge to simulate long-term deterioration of the neutralizing compounds of the gas filter. Then perform a GB challenge to determine the remaining absorption capacity of the filter media. Record the following:

a. Concentration and time duration of each agent at the protector inlet.

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b. Flow rate of air into the installation.
c. Contamination of air inside the installation.
d. Time and concentration of CK required to break down the filters.

6.2.10.3 Purge Test

The ability of the test item to purge the protective installation of contaminated air will be tested as follows:

a. Contaminate the simulated installation with the agent(s) specified by the item's design requirements or the test plan.
b. Operate the test item so as to recirculate 100 percent of the air in the installation.
c. Record the following:

1) Volume of installation.
2) Contaminant used.
3) Concentration of contaminant.
4) Reduction of contamination as a function of test item operating time.
5) Baffle arrangement to direct airflow inside the installation.

d. Repeat steps a through c with other contaminants or different baffle arrangement in the installation as directed by the test plan.

6.2.11 Decontamination Aspects

Decontaminate the test item as described in the applicable portions of MTP 8-2-510 and record the following:

a. Decontamination and methods which provide the most satisfactory results.
b. The effects of the most suitable decontaminant on test items.
c. Safety precautions used to protect personnel from decontaminants.
d. Ease of performing decontaminating operations at night.
e. The operability of test item after decontamination as measured by the applicable procedures of paragraphs 6.2.9 and 6.2.10.

6.2.12 Maintenance Aspects

NOTE: Background information on checking a test item to determine its maintenance aspects is contained in reference 4B (USAMC Pamphlet 706-134).

a. Inspect the test item for deficiencies which will require replacement of components before the item can be tested. Photograph all deficiencies.
b. Complete authorized maintenance tasks in accordance with the test item maintenance manuals, manufacturer's instructions, etc.
c. Record the time and personnel required to perform scheduled and

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nonscheduled maintenance tasks throughout the period of testing.

d. Determine the accuracy and evaluate the adequacy of the test item maintenance package.

e. Note whether special tools or skills are required.

f. Record the following:

1) Test item down time (cumulative)
2) Time taken between repairs and reasons, if appropriate
3) Frequency of repairs
4) Nomenclature of repair parts used

g. Evaluate the test design from maintainability standpoint, as applicable.

6.2.13 Human Factors Aspects

Throughout the conduct of all tests required by this MTP, observations will be made regarding the human factors aspects of the test item. Record all inconvenience encountered in handling and operating the test item and any accompanying instructional materials, including the following:

NOTE: References 4D (MIL-STD-1472) and reference 4E (MIL-H-46855) for discussions of the human factors relevant to the items under test.

a. Compatibility with field clothing and equipment, particularly with protective clothing.

b. Simplicity and adequacy of operating instructions.

c. Conformance of the test item's design to qualitative requirements, particularly whether it is as quiet and compact as possible commensurate with functional characteristics.

d. Ease of identifying items and components under daylight, darkness, and blackout conditions.

6.3 TEST DATA

6.3.1 Receipt Inspection

a. Record the data collected as described in applicable sections of MTP 8-2-500 and the following:

1) For the test item package:

a) Binding of packing material deficiencies.
b) Indications of damage, deterioration, or illegible markings.
c) Length, width, height, in feet and inches.
d) Evidence of rust or corrosion.
e) Presence of packing lists and uncrating instructions.
f) Weight, in pounds.

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g) Missing packages.
h) Adequacy of unpacking instructions.
i) Special tools required for unpacking.

2) For the test item:

a) Evidence of damage or deterioration
b) Length, maximum height and diameter, in feet and inches
c) Weight, in pounds
d) Type
e) Number of filters per package

b. Retain all photographs.

6.3.2 Installation

Record the following:

a. Missing components.
b. Items which have improper fit.
c. Difficulty in assembling the test item.
d. Ease of installing the filter in the collective protection system.

6.3.3 Safety Evaluation

Record the following:

a. Any hazardous characteristics
b. Test item deficiencies and recommendations for improvement
c. Information for inclusion in the Safety Release Recommendation

6.3.4 Simulated Environmental Testing

6.3.4.1 Cyclic Storage

Record the following for each cycle:

a. Test item identification number
b. Cycle number
c. Damage to:

1) Container
2) Test item

6.3.4.2 Extreme Temperature Tests

6.3.4.2.1 Low Temperature Tests -

Record the following for each test item, as applicable:

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a. Test item identification number
b. For temperature of -53.9°C (-65°F); damage incurred
c. For minimum operating temperature:

1) Temperature in °C and °F.
2) Damage incurred.
3) Leakage test data collected as described in paragraph 6.2.8.
4) Operability data collected as described in paragraphs 6.2.9 and 6.2.10.

d. For ambient temperature:

1) Temperature in °C and °F.
2) Test item damage.
3) Leakage data collected as described in paragraph 6.2.8.
4) Operability data collected as described in paragraphs 6.2.9 and 6.2.10.

6.3.4.2.2 High Temperature -

Record the following for each test item, as applicable:

a. Test item identification number
b. For temperature of 68.3°C (155°F) damage incurred
c. For temperature of 48.9°C (120°F):

1) Damage incurred.
2) Leakage data collected as described in paragraph 6.2.8.
3) Operability data collected as described in paragraph 6.2.9 and 6.2.10.

d. For ambient temperature:

1) Temperature in °C and °F.
2) Relative humidity in % .
3) Damage incurred.
4) Leakage data collected as described in paragraph 6.2.8.
5) Operability data collected as described in paragraph 6.2.9 and 6.2.10.

6.3.4.3 Fungus Test

Record the following for each test item:

a. Test item identification number
b. Operability data collected as described in paragraph 6.2.10
c. Presence of fungus on test item and components
d. Evidence of corrosion

6.3.4.4 Humidity Test

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Record the following for each test item:

a. Test item identification number.
b. Operability data collected as described in paragraph 6.2.9 and 6.2.10.
c. Evidence of corrosion on test item and components.

6.3.4.5 Dust Test

Record the following for each test item:

a. Test item identification number.
b. Operability data collected as described in paragraph 6.2.9 and 6.2.10.
c. Damage to:

1) External surfaces
2) Test item components

d. Presence of dust on test item components

6.3.4.6 Water Immersion Test

Record the following for each test item as applicable:

a. Test item identification number
b. During immersion:

1) Presence of bubbling, if any
2) Immersion time to bubbling, if any, in minutes

c. Presence of water penetration.
d. Operability data collected as described in paragraph 6.2.9 and 6.2.10.

6.3.4.7 Salt Fog Test

Record the following for each test item as applicable:

a. Test item identification number.
b. Operability data collected as described in paragraphs 6.2.9 and 6.2.10.
c. Evidence of corrosion on test item and components.
d. Evidence of water penetration.
e. Leakage data collected as described in paragraph 6.2.8.

6.3.4.8 Rain Test

Record the following for each test item, as applicable:

a. Test item identification number

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b. Presence of corrosion:

1) Test item
2) Test item components

c. Evidence of water penetration

6.3.5 Rough Handling and Surface Transport Tests

Record the following for each test item, as applicable:

a. Test item identification number.
b. Data collected as described in the applicable sections of MTP 8-2-503.
c. Leakage data collected as described in paragraph 6.2.8.
d. Operability data collected as described in paragraph 6.2.9 and 6.2.10.

6.3.6 Air Transportability

Record the following:

a. Data collected as described in the applicable sections of MTP 7-2-515.
b. Type of aircraft used or simulated.
c. Shipping container:

1) Length, width, and height, in inches
2) Weight, in pounds
3) Material

d. Equipment used in loading
e. Difficulties encountered while loading
f. Method of tiedown
g. Damage incurred to the package while loading
h. Equipment used in unloading
i. Difficulties incurred in unloading
j. Damage incurred to the package while unloading

6.3.7 Airdrop Capability

a. Record the following for each test item:

1) Test item identification
2) Type aircraft used
3) Aircraft altitude in feet
4) Aircraft airspeed in mph
5) Air conditions (calm, turbulent)
6) For meteorological conditions:

a) Temperature in °C

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b) Wind speed in mph
c) Wind direction
d) Relative humidity in percent
e) Barometric pressure

7) Deceleration magnitude at impact in g's
8) Impact velocity in fps
9) For test item package:

a) Presence of cracks, breaks, etc
b) Undone binding

10) For air test item:

a) Damage or deformities.
b) Leakage data collected as described in paragraph 6.2.8.
c) Operability data collected as described in paragraphs 6.2.9 and 6.2.10.

b. Retain all motion and still pictures

6.3.8 Leak Testing

Record data collected as described in the applicable sections of MTP 8-2-511.

6.3.9 Filter Tests

6.3.9.1 Flow-Rate Test

Record the following:

a. Test item identification number (filter serial number)
b. Airflow rate in cfm
c. Pressure drop across filter in psi
d. Testing difficulties, if any

6.3.9.2 Dioctyl Phthalate (DOP) Filter Penetration Test

Record the following:

a. Test item identification number (filter serial number)
b. Type, size, and weight of filter
c. Flow rate of air in cfm
d. Pressure drop across filter
e. DOP particle size range
f. DOP density in purified air
g. Reasons for deviation from specifications, if known

6.3.9.3 Filter Gas Life Test

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Record the following:

a. Test item identification number (filter serial number)
b. Air flow rate in cfm
c. Flow resistance of filters in inches of water
d. Type and concentration of agent used
e. Density of agent in the purified air
f. Filter life
g. Weight of filters before and after test

6.3.10 Operational Reliability

6.3.10.1 Flow-Rate Test

Record the following:

a. Test item identification number
b. Portion of air which is recirculated
c. Air flow rate in cfm
d. Static air pressure in the shelter in psi
e. Stall pressure of the test item in psi

6.3.10.2 Agent Challenge Tests

6.3.10.2.1 Dioctyl Phthalate (DOP) Challenge -

Record the following:

a. Test item identification number.
b. Rate of agent penetration of the collective protector.
c. Agent concentration in the protective installation measured as as function of time.
d. Air flow rate in cfm
e. Static pressure in shelter in psi

6.3.10.2.2 Sarin (GB) Challenge -

Record data collected as described in paragraph 6.2.10.2.1.

6.3.10.2.3 Distilled Mustard Gas (HD) Challenge -

Record data collected as described in paragraph 6.2.10.2.1.

6.3.10.2.4 Cyanogen Chloride (CK) and Sarin (GB) Challenge -

Record the following:

a. Concentration and time duration of challenge for each agent
b. Flow rate of air into the installation
c. Level of contamination of air inside the installation
d. Time and concentration of CK required to break down the filters

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6.3.10.3 Purge Test

Record the following:

a. Volume of installation.
b. Contaminant used.
c. Concentration of contaminant.
d. Reduction of contamination as a function of test item operation time.
e. Arrangement of baffles inside the installation to direct air flow.

6.3.11 Decontamination Aspects

Record the following for each test item undergoing decontamination:

a. Data collected as described in the applicable sections of MTP 8-2-510.
b. Operability data collected as described in the applicable procedures of paragraph 6.2.9 and 6.2.10.

6.3.12 Maintenance Characteristics

a. Record the following:

1) Adequacy and accuracy of maintenance package:

a) Scheduled maintenance
b) Unscheduled maintenance

2) Time required to perform, in minutes:

a) Scheduled maintenance
b) Unscheduled maintenance

3) Personnel required to perform:

a) Scheduled maintenance
b) Unscheduled maintenance

4) Cumulative down time in hours
5) Time taken between repairs in hours, if any
6) Reason for delay in repairs, if any
7) Frequency of repairs
8) Nomenclature of parts used
9) Special tools required
10) Special skills required

b. Retain all photographs

6.3.13 Human Factors

Record the following:

a. Compatibility with field clothing and equipment.
b. Adequacy of operating instructions.

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c. Conformance to qualitative requirements.

d. Ease of identifying items and components under daylight, darkness, and blackout conditions.

6.4 DATA REDUCTION AND PRESENTATION

6.4.1 Receipt Inspection

a. Data collected as a result of this procedure shall be presented as indicated in the applicable sections of MTP 8-2-500.

b. The description of the test item, number of items tested, and conditions upon receipt shall be presented in tabular form.

c. Results of the leak subtest shall be presented in narrative or other convenient form.

d. Photographs shall be used to substantiate results.

6.4.2 Installation

a. The results of this procedure shall be presented in tabular or other suitable form.

b. Photographs shall be used to show the configuration of the test set-up.

6.4.3 Safety Evaluation

a. A Safety Release Recommendation (USATECOM Regulation 385-6) shall be forwarded to the U. S. Army Test and Evaluation Command within 30 days of the beginning of the test. The Safety Release Recommendation shall contain the following information: special safety considerations or hazards to personnel and materiel (including developmental types of equipment as well as standard components used in assemblage of items being tested).

b. Data and comments relative to the safety hazards observed during any phase of testing.

c. Comments relative to suggested safety improvements.

6.4.4 Simulated Environmental Testing

a. The results of the subtests conducted shall be presented in tabular or other suitable form.

b. The results of the operational check tests performed at the conclusion of the various environmental tests shall be presented in narrative or other suitable form.

6.4.5 Rough Handling and Surface Transport

a. Rough Handling and surface transport data shall be presented as prescribed in MTP 8-2-503.

b. Vibration and shock data shall be presented in tabular form to indicate test times, distances(dropped), shock levels, vibration frequencies, etc., and significant findings of the test. Include photographs of damage.

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c. Present data on operation of test item after subjection to rough handling and surface transport, conditions, vibration and shock.

6.4.6 Air Transportability

a. Data shall be presented in summary form as indicated in the applicable sections of MTP 7-2-515, and other pertinent testing documentation.

b. Present data regarding any significant aspects of the test item observed during conduct of air transport testing.

6.4.7 Airdrop Capability

a. The results of the subtest shall be presented as prescribed in MTP 7-2-509 and include the following:

1) Type of aircraft
2) Air speed, altitude, and meteorological conditions
3) Packaging material condition after test
4) Maximum "G" force on opening of parachute and on impact

b. Present narrative comments and data regarding ease or difficulty encountered in accomplishing airdrop. Present photographs (as required) to indicate results of airdrop.

c. Present data on operation and performance of the test item after airdrop capability subtest.

6.4.8 Leak Testing

a. The results of leak testing shall be presented as prescribed in MTP 8-2-511.

b. Narrative comments, photos, etc., shall be included, as required.

6.4.9 Filter Tests

a. For each CBR agent used prepare graphs showing the following:

1) Airflow rate vs. time
2) Pressure drop across the filter vs. time
3) Purity of filtered air vs. time

b. Present all other data derived in narrative form supplemented by drawings, photographs, charts, tables, graphs or any other suitable means of displaying the results of the test.

c. The report shall clearly conclude whether the test items meet the reliability criteria established in the applicable specifications. Recommendations relative to further and future testing and methods to overcome malfunctions shall also be included.

6.4.10 Operational Reliability

a. Data derived from this subtest shall be presented in narrative form,

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supplemented by drawings, photographs, charts, tables, graphs or any other suitable means of displaying information. Data evaluation shall specifically include but not be limited to the following:

1) Air Flow Rate
2) Static pressure within the shelter
3) Percent of air recirculated
4) Purity of filtered air
5) Effects of different challenging agents on the unit
6) Mean time required to purge the unit
7) Ease of operation

b. The report shall clearly conclude whether the test item meets the reliability criteria established in applicable specifications. Recommendations relative to further testing and methods to overcome malfunctions shall also be included.

6.4.11 Decontamination Aspects

The results of this subtest shall be presented as indicated in the applicable sections of MTP 8-2-510.

6.4.12 Maintenance Aspects

Data from this subtest shall be presented in narrative form. The report shall be supplemented by photos, drawings, or other devices or substantiate the conclusions and recommendations.

6.4.13 Human Factors

a. Data from this subtest shall be presented in tabular, narrative, or other suitable form supplemented by photographs and graphic or art presentations as required.

b. A summary of comments regarding shortcomings and recommended improvements shall be presented.

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